

Knowledge Base (CC BY-SA 4.0)

Machine Learning on Knowledge Bases

Rule Mining Algorithm (2)

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Previous Homework

Question: Calculate Support and PCA confidence scores for the following rule.

Rule: $livesIn(x, y) \Rightarrow isPoliticianOf(x, y)$

KB:

<i>livesIn</i>	<i>isPoliticianOf</i>	<i>diedIn</i>
(Ava, Paris)	(Ava, Paris)	(Ava, Paris)
(Emily, London)	(Bob, Newyork)	(Bob, Newyork)
(Emily, Paris)	(Ava, Newyork)	(Emma, Lisbon)

Support

Support: the number of facts predicted by R that exist in $\mathcal{K}$

$$\text{support}(R) := |\{p : (\mathcal{K} \wedge R \models p) \wedge p \in \mathcal{K}\}|$$

Rule: $\text{livesIn}(x, y) \Rightarrow \text{isPoliticianOf}(x, y)$

The rule generates three facts:

- ▶ $\text{isPoliticianOf}(\text{Ava}, \text{Paris})$
- ▶ $\text{isPoliticianOf}(\text{Emily}, \text{London})$
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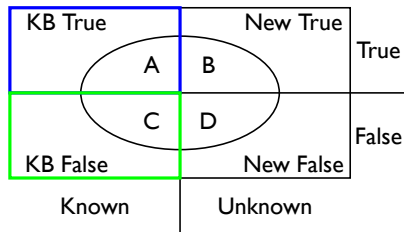
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$$\underline{\text{support}(\text{livesIn}(x, y) \Rightarrow \text{isPoliticianOf}(x, y)) = 1}$$

PCA Confidence

The central challenge is to provide counter-examples (negative samples) for the rule mining



Confidence: $cex(R)$ is a set of counter examples for a rule R

$$conf(R) := \frac{support(R)}{support(R) + \underbrace{|\{p : (\mathcal{K} \wedge R \models p) \wedge p \in cex(R)\}|}_{\text{The number of false predictions}}}$$

Partial Completeness Assumption (PCA): whenever at least one object for a given subject and a relation is in a KB, then all objects for that subject-relation pair are assumed to be known

PCA confidence:

$$\begin{aligned} \text{conf}_{pca}(R) &= \frac{\text{support}(R)}{\text{support}(R) + |(x, y) : (\mathcal{K} \wedge R \models r(x, y)) \wedge r(x, y') \in \mathcal{K} \wedge r(x, y) \notin \mathcal{K}|} \\ &= \frac{\text{support}(R)}{|(x, y) : (\mathcal{K} \wedge R \models r(x, y)) \wedge r(x, y') \in \mathcal{K}|} \end{aligned}$$

PCA confidence: $conf_{pca}(R) = \frac{support(R)}{|(x,y):(\mathcal{K} \wedge R \models r(x,y)) \wedge r(x,y') \in \mathcal{K}|}$

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$isPoliticianOf(Emily, *)$ does not exist in $\mathcal{K}$

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$isPoliticianOf(Emily, *)$ does not exist in $\mathcal{K}$

$conf_{pca}(livesIn(x,y) \Rightarrow isPoliticianOf(x,y)) = \frac{1}{1} = 1$

$$\mathbf{PCA\ confidence:} \text{ conf}_{pca}(R) = \frac{\text{support}(R)}{|(x,y):(\mathcal{K} \wedge R \models r(x,y)) \wedge r(x,y') \in \mathcal{K}|}$$

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$\text{isPoliticianOf}(\text{Emily}, *)$ does not exist in $\mathcal{K}$

$$\text{conf}_{pca}(\text{livesIn}(x,y) \Rightarrow \text{isPoliticianOf}(x,y)) = \frac{1}{1} = 1$$

but $\text{isPoliticianOf}(\text{Ava}, *)$ exists in $\mathcal{K}$, so $\text{conf}_{pca}(R) = \frac{1}{2}$ is also correct

$$\text{conf}_{pca}(\text{livesIn}(x,y) \Rightarrow \text{isPoliticianOf}(x,y)) = 1$$

- ▶ This means that a human x who lives in y is always a politician of y
- ▶ Is this true ? $\Rightarrow$ No

Problem of PCA

PCA relies on the fact that relations in KBs tend to be *functional*

- ▶ e.g.: The relation *hasBirthday*(x, y) is function (*hasBirthday*(Russell, 18_May_1872))

However, the assumption may be violated whenever a relation may associate multiple objects with each subject

Several papers address the above problem:

- ▶ “Completeness-aware Rule Learning from Knowledge Graphs”, Pellissier et. al., IJCAI, 2018.
- ▶ “Estimating Rule Quality for Knowledge Base Completion with the Relationship between Coverage Assumption”, Zupanc and Davis, WWW, 2018.

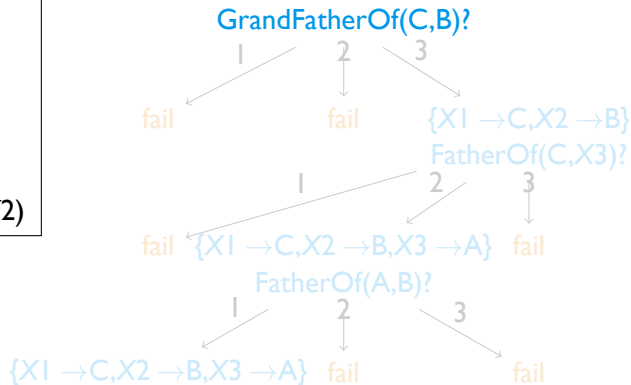
TODAY'S AGENDA

- ▶ Review What is Rule Mining
- ▶ Coverage Assumption

Logical Deductive Inference

KB Example:

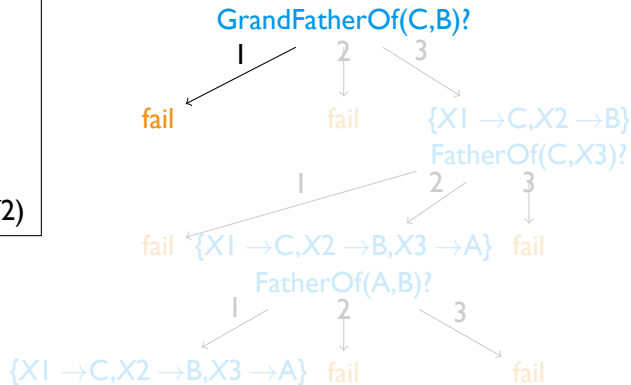
1. $\text{FatherOf}(A,B)$
2. $\text{FatherOf}(C,A)$
3. $\text{FatherOf}(X1,X3) \wedge \text{FatherOf}(X3,X2) \Rightarrow \text{GrandFatherOf}(X1,X2)$



Logical Deductive Inference

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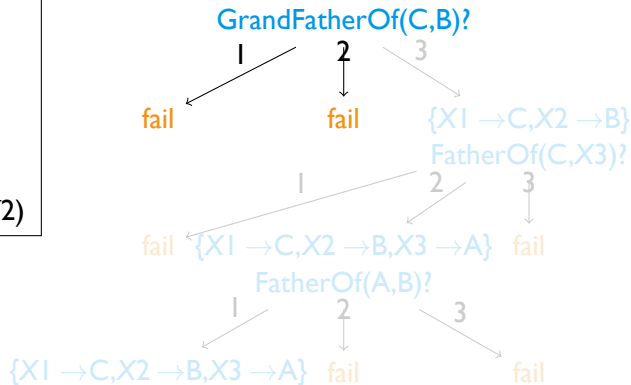
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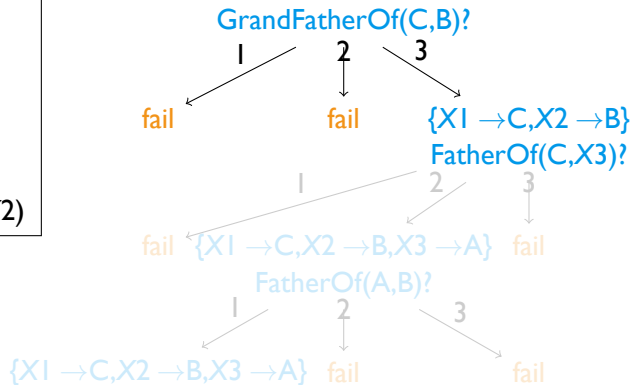
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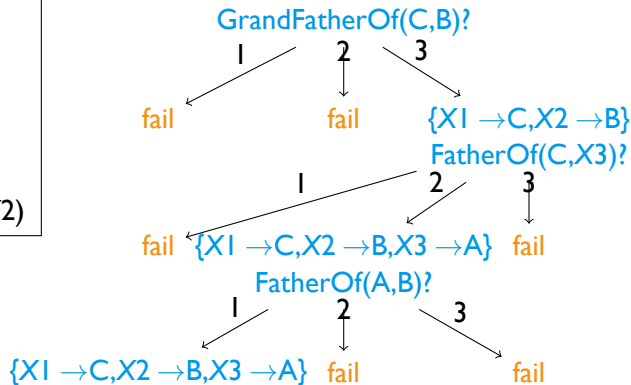
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Atoms and Horn Rules

Atom: a fact that can have variables at the subject and/or object position

Rule Example:

$$hasChild(X1, X2) \wedge isCitizenOf(X1, X3) \Rightarrow isCitizenOf(X2, X3)$$

Horn Rule: a rule with Body and Head by implication

$$\underbrace{B_1 \wedge B_2 \wedge \cdots \wedge B_n}_{\text{Body}} \Rightarrow \underbrace{r(X, Y)}_{\text{Head}}$$

where Body is a set of Atoms and Head is a single atom (Abbreviation $\vec{B} \Rightarrow r(X, Y)$)

Instantiation and Prediction

Instantiation of a rule: a copy of the rule where all variables have been substituted by constants

- ▶ e.g.: An instantiation of a rule $\text{livesIn}(x, y) \Rightarrow \text{wasBornIn}(x, y)$ is $\text{livesIn}(\text{Adam}, \text{Paris}) \Rightarrow \text{wasBornIn}(\text{Adam}, \text{Paris})$

Prediction of a rule: the head atom of an instantiated rule if all body atoms of the instantiated rule appear in the KB $\mathcal{K}$

- ▶ Notation: $\mathcal{K} \wedge R \models p$
- ▶ e.g.: The prediction of the instantiated rule $\text{livesIn}(\text{Adam}, \text{Paris}) \Rightarrow \text{wasBornIn}(\text{Adam}, \text{Paris})$ is the head atom $\text{wasBornIn}(\text{Adam}, \text{Paris})$ ($\text{livesIn}(\text{Adam}, \text{Paris}) \in \mathcal{K}$)

The Aim of Rule Mining

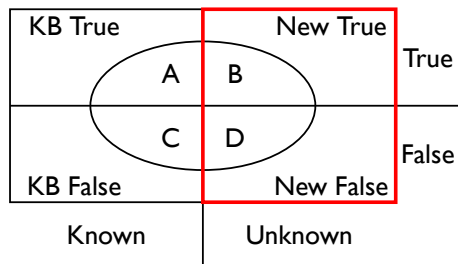
KB True	New True	True
A	B	Predictions
KB False	New False	
C	D	False
Known	Unknown	

1. KB True: True facts that are known to the KB
2. New True: True facts that unknown to the KB
3. KB False: Facts that are known to be false in the KB
4. New False: Facts that are false but unknown to the KB

The aim of rule mining is to find rules that make true predictions

- ▶ maximize the area B and minimize the area D
- ▶ B and D are unknown and we need to design good measures of rules

The Aim of Rule Mining



1. KB True: True facts that are known to the KB
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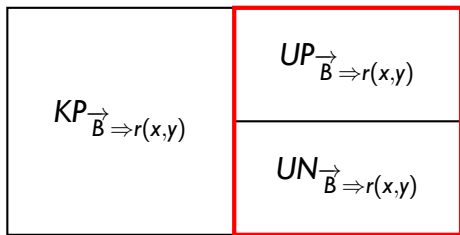
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TODAY'S AGENDA

- ▶ Review What is Rule Mining
- ▶ Coverage Assumption

Redefinition: Confidence (1/2)

$$P_{\vec{B} \Rightarrow r(x,y)}$$



$$U_{\vec{B} \Rightarrow r(x,y)}$$

$$P_{\vec{B} \Rightarrow r(x,y)}:$$

- ▶ the set of predictions made by a rule $\vec{B} \Rightarrow r(x,y)$

Each prediction made by a rule $\vec{B} \Rightarrow r(x,y)$ either falls in the set of known positive examples $KP_{\vec{B} \Rightarrow r(x,y)}$ or belongs to a set $U_{\vec{B} \Rightarrow r(x,y)}$ of unlabeled examples

Unknown examples can be subdivided into two sets:

- ▶ **Unk. Positives** $UP_{\vec{B} \Rightarrow r(x,y)}$
- ▶ **Unk. Negatives** $UN_{\vec{B} \Rightarrow r(x,y)}$

Redefinition: Confidence (2/2)

$P_{\vec{B} \Rightarrow r(x,y)}$

$KP_{\vec{B} \Rightarrow r(x,y)}$	$UP_{\vec{B} \Rightarrow r(x,y)}$
	$UN_{\vec{B} \Rightarrow r(x,y)}$

$U_{\vec{B} \Rightarrow r(x,y)}$

Confidence $conf(\vec{B} \Rightarrow r(x,y))$:

$$\frac{\text{support}(\vec{B} \Rightarrow r(x,y)) + |UP_{\vec{B} \Rightarrow r(x,y)}|}{|P_{\vec{B} \Rightarrow r(x,y)}|}$$

Key Insight: estimating a rule's confidence only requires knowing the size of the set $UP_{\vec{B} \Rightarrow r(x,y)}$

How to Estimate $|UP_{\vec{B} \Rightarrow r(x,y)}|$

Confidence:

$$conf(\vec{B} \Rightarrow r(x,y)) = \frac{support(\vec{B} \Rightarrow r(x,y)) + |UP_{\vec{B} \Rightarrow r(x,y)}|}{|P_{\vec{B} \Rightarrow r(x,y)}|}$$

Two key questions:

1. What is the relationship between the proportion of known and unknown positives covered by a rule ?
2. What percentage of the unlabeled data for a given relation is expected to be positive ?

Coverage Assumption (1/2)

Question 1.:

- What is the relationship between the proportion of known and unknown positives covered by a rule ?

Answer $\Rightarrow$ **Coverage Assumption:**

- The proportion of positive examples covered by a rule is the same for both the labeled and unlabeled examples

$$rc(\vec{B} \Rightarrow r(x,y)) = \frac{\text{support}(\vec{B} \Rightarrow r(x,y))}{|\mathcal{K}_r|} = \frac{|UP_{\vec{B} \Rightarrow r(x,y)}|}{|UP_r|}$$

where $\mathcal{K}_r$ is the set of all facts for relation r that appear in $\mathcal{K}$ and UP_r represents the set of all true facts for relation r that do not appear in $\mathcal{K}$

Coverage Assumption (2/2)

Coverage Assumption:

$$rc(\vec{B} \Rightarrow r(x,y)) = \frac{\text{support}(\vec{B} \Rightarrow r(x,y))}{|\mathcal{K}_r|} = \frac{|UP_{\vec{B} \Rightarrow r(x,y)}|}{|UP_r|}$$

We can compute $rc(\vec{B} \Rightarrow r(x,y))$ because

- we know $\text{support}(\vec{B} \Rightarrow r(x,y))$ and $\mathcal{K}_r$

If we know $|UP_r|$, we could estimate $|UP_{\vec{B} \Rightarrow r(x,y)}|$

How to Estimate $|UP_r|$ (1/7)

Question 2.:

- What percentage of the unlabeled data for a given relation is expected to be positive ?

We use β to denote the proportion of all the unlabeled examples that belong to the positive class

$$\beta = \frac{|UP_r|}{|U_r|}$$

where U_r represents the set of all facts for relation r that do not appear in $\mathcal{K}$

Answer $\Rightarrow$ Computing β is somewhat tricky

How to Estimate $|UP_r|$ (2/7)

PCA: whenever at least one object for a given subject and a relation is in a KB, then all objects for that subject-relation pair are assumed to be known

$$\beta_{PCA} = \frac{|\mathcal{K}_r|}{|\mathcal{S}_r| \times |\mathcal{O}_r|}$$

where $\mathcal{S}_r$ and $\mathcal{O}_r$ denote the set of constants that appear as a subject or object in relation r in $\mathcal{K}$

How to Estimate $|UP_r|$ (3/7)

A better possibility is to define β on a per-rule basis at:

$$\beta_{\vec{B} \Rightarrow r(x,y)} = \frac{\text{support}(\vec{B} \Rightarrow r(x,y))}{|\mathcal{S}_{\vec{B} \Rightarrow r(x,y)}| \times |\mathcal{O}_{\vec{B} \Rightarrow r(x,y)}|}$$

where

$$\begin{aligned}\mathcal{S}_{\vec{B} \Rightarrow r(x,y)} &= |\{x \mid \exists x : r(x,y) \in P_{\vec{B} \Rightarrow r(x,y)}\}| \\ \mathcal{O}_{\vec{B} \Rightarrow r(x,y)} &= |\{y \mid \exists y : r(x,y) \in P_{\vec{B} \Rightarrow r(x,y)}\}| \end{aligned}$$

How to Estimate $|UP_r|$ (4/7)

β : the proportion of all the unlabeled examples that belong to the positive class

Our goal is to compute $|UP_r|$:

$$|UP_r| = \beta_{\vec{B} \Rightarrow r(x,y)} \times \underbrace{|\{r(x,y) \mid x \in \mathcal{S}_{\vec{B} \Rightarrow r(x,y)}, y \in \mathcal{O}_{\vec{B} \Rightarrow r(x,y)}\} \setminus \mathcal{K}_r|}_{|U_{\vec{B} \Rightarrow r(x,y)}|}$$

However, this ignores the subtlty that each relation behaves differently in terms of the number of objects we would expect to be associated with each subject and vice versa

How to Estimate $|UP_r|$ (5/7)

Consider partitioning $U_{\vec{B} \Rightarrow r(x,y)}$ into the following four subsets:

$$S_{\text{old}}O_{\text{old}} = \{r(x,y) \mid x \in (\mathcal{S}_{\vec{B} \Rightarrow r(x,y)} \cap \mathcal{S}_r), y \in (\mathcal{O}_{\vec{B} \Rightarrow r(x,y)} \cap \mathcal{O}_r)\} \setminus \mathcal{K}_r$$

$$S_{\text{old}}O_{\text{new}} = \{r(x,y) \mid x \in (\mathcal{S}_{\vec{B} \Rightarrow r(x,y)} \cap \mathcal{S}_r), y \in (\mathcal{O}_{\vec{B} \Rightarrow r(x,y)} \setminus \mathcal{O}_r)\}$$

$$S_{\text{new}}O_{\text{old}} = \{r(x,y) \mid x \in (\mathcal{S}_{\vec{B} \Rightarrow r(x,y)} \setminus \mathcal{S}_r), y \in (\mathcal{O}_{\vec{B} \Rightarrow r(x,y)} \cap \mathcal{O}_r)\}$$

$$S_{\text{new}}O_{\text{new}} = \{r(x,y) \mid x \in (\mathcal{S}_{\vec{B} \Rightarrow r(x,y)} \setminus \mathcal{S}_r), y \in (\mathcal{O}_{\vec{B} \Rightarrow r(x,y)} \setminus \mathcal{O}_r)\}$$

Like PCA, we ignore $S_{\text{old}}O_{\text{old}}$ and assume that all of these predictions are false

How to Estimate $|UP_r|$ (6/7)

We scale back the estimated number of positive examples in $S_{\text{old}}O_{\text{new}}$ and $S_{\text{new}}O_{\text{old}}$

- We need to account for the fact that each subject already appears in $\mathcal{K}_r$ with some objects, so we would expect it to associate with fewer new objects than if the subject did not appear in $\mathcal{K}_r$

We use **functionality** and its inverse functions (if r is function (inverse function), they become zero):

$$f_S(r) = 1 - HM_{x \in \mathcal{S}_r} \left(\frac{1}{|\{y \mid r(x, y) \in \mathcal{K}\}|} \right)$$
$$f_O(r) = 1 - HM_{x \in \mathcal{O}_r} \left(\frac{1}{|\{x \mid r(x, y) \in \mathcal{K}\}|} \right)$$

How to Estimate $|UP_r|$ (7/7)

Finally, we define $|UP_r|$ as follows:

$$|UP_{S_{old}O_{new}}| = |U_{S_{old}O_{new}}| \times \beta_{\vec{B} \Rightarrow r(x,y)} \times f_S(r) \times rc(\vec{B} \Rightarrow r(x,y))$$

$$|UP_{S_{new}O_{old}}| = |U_{S_{new}O_{old}}| \times \beta_{\vec{B} \Rightarrow r(x,y)} \times f_O(r) \times rc(\vec{B} \Rightarrow r(x,y))$$

$$|UP_{S_{new}O_{new}}| = |U_{S_{new}O_{new}}| \times \beta_{\vec{B} \Rightarrow r(x,y)} \times rc(\vec{B} \Rightarrow r(x,y))$$

where $rc(\vec{B} \Rightarrow r(x,y))$ represents the proportion of positive examples covered by a rule is the same for both the labeled and unlabeled examples

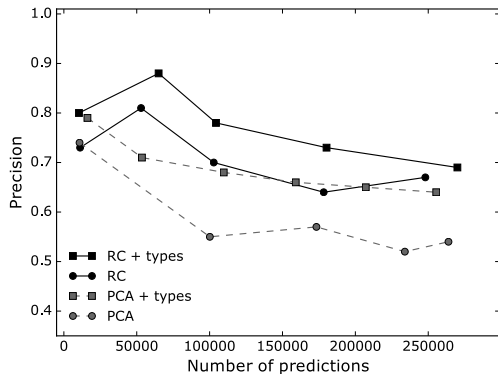
$$rc(\vec{B} \Rightarrow r(x,y)) = \frac{support(\vec{B} \Rightarrow r(x,y))}{|\mathcal{K}_r|} = \frac{|UP_{\vec{B} \Rightarrow r(x,y)}|}{|UP_r|}$$

Confidence over Coverage Assumption

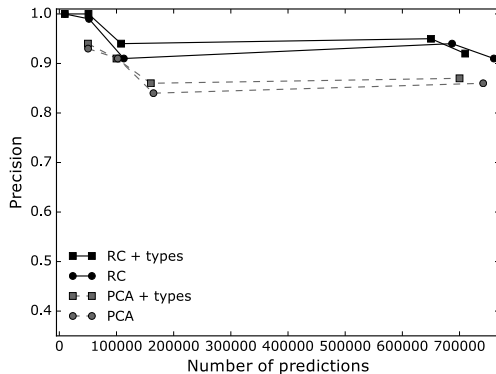
Confidence:

$$\begin{aligned} \text{conf}(\vec{B} \Rightarrow r(x,y)) &= \frac{\text{support}(\vec{B} \Rightarrow r(x,y)) + |UP_{\vec{B} \Rightarrow r(x,y)}|}{|P_{\vec{B} \Rightarrow r(x,y)}|} \\ &= \frac{\text{support}(\vec{B} \Rightarrow r(x,y)) + |UP_{s_{\text{old}}o_{\text{new}}}| + |UP_{s_{\text{new}}o_{\text{old}}}| + |UP_{s_{\text{new}}o_{\text{new}}}|}{|P_{\vec{B} \Rightarrow r(x,y)}|} \end{aligned}$$

Rule Evaluation



(a) YAGO2



(b) Wikidata